# MSCL: Multi-stage Curriculum Learning for Postoperative Glioma Segmentation Based on Fine-tuning Foundation Model

Anonymized Authors

Anonymized Affiliations
email@anonymized.com

**Abstract.** Precise segmentation of postoperative glioma poses a pivotal task in medical image processing due to the crucialness of assessing treatment protocols and directing further therapeutic measures. Despite numerous studies on primary tumors and the existence of highly effective models, research on residual tumors is relatively scarce due to difficulties in annotation and sample acquisition. Since preoperative glioma are tumors before surgical resection of postoperative glioma, providing significant assistance in the segmentation of gliomas postsurgery, we propose a curriculum learning approach for postoperative glioma segmentation. Instead of directly segmenting postoperative glioma, our method establishes a multistage curriculum system based on the ease of data, enabling the model to engage in continuous learning from preoperative to postoperative data, which preserves the dark knowledge acquired in previous tasks. We train the public dataset of primary tumors at the first stage for easy data learning and postoperative glioma public dataset at the second stage for medium data learning, ultimately, we undertake target domain adaptation for background knowledge learning at the last stage for hard data learning. We evaluated the proposed method on three public postoperative glioma datasets and one private dataset, achieving improvements in DSC by 11.7%, 15.1%, 8.2%, and 14.2% compared to fine-tuning foundation model directly; 11.4%, 7.8%, 14.5% and 8.7% compared to state-of-the-art methods.

**Keywords:** Curriculum Learning · Foundation Model · Medical Image Segmentation.

## 1 Introduction

Glioma represents the most prevalent type of primary intracranial tumor, stemming from the malignant transformation of glial cells in the brain and spinal cord. Surgical resection is the primary choice of treatment for glioma. However, due to the location and invasive nature of the tumour, complete resection can be challenging, mainly when the tumour is situated in functional areas of the brain. Therefore, doctors should develop optimal treatment strategies based on the patient's post-operative condition to effectively control tumour growth and

improve prognosis. It underscores the importance of accurately identifying residual tumours after surgery. Nevertheless, the identification of residual tumours is a very challenging task. Because of the inherent infiltrative growth pattern of gliomas, there is no clear boundary between the residual tumor and adjacent tissues, and the tumor tissues become fragmented after further surgery, all of which pose a great challenge in the segmentation of the residual tumor.

To effectively address the challenges posed by these complex and diverse data, Curriculum Learning, as a gradual learning strategy, is adopted in our method. In this paper, we establish a multi-stage curriculum system based on the complexity of the tumor, enabling the model to progressively learn from simple to complex knowledge and to continuously learn from preoperative to postoperative data, thereby retaining the tacit knowledge acquired from previous tasks. In the first stage, we incorporate publicly available datasets of smaller primary tumors into the training set, allowing the model to acquire the ability to segment primary tumors. In the second stage, we introduce datasets of residual tumors with gold standards, enabling the model to learn a certain level of ability to segment residual tumors. However, due to differences in background knowledge between public datasets and the target domain, applying the model from this stage to segment residual tumors in the target domain may not yield good results. Therefore, we incorporate a third stage into the curriculum for target domain adaptation focused on background knowledge learning. Here we propose data simulation approaches for residual tumour segmentation, utilizing the preoperative dataset to simulate surgical resection and generate simulated postoperative residual tumours adding into the thrid stage training set. This method obtains precise boundaries and gold-standard masks of simulated post-operative tumors. For summarization, our major contributions are:

1) We have established a multi-stage curriculum system based on the complexity of tumors, enabling the model to progressively learn from simple to complex knowledge. It continuously learns from preoperative to postoperative data, ultimately achieving excellent segmentation performance for postoperative gliomas in the target domain.

2) We have proposed a data simulation model that utilizes the preoperative dataset to simulate resection operations, generating simulated diverse-volume postoperative gliomas for target domain background adaptation.

3) We have designed a fine-tuning strategy that is combined with the optimization of prompt points in the last stage, which integrates a novel loss function that incorporates the MSE loss between the grayscale threshold of the predicted tumor region, calculated through hint points, and the gray-scale threshold of the actual tumor region into the loss of segmentation training. This integration aims to optimize the prompt points and achieve precise segmentation.

## 2 Methodology

The Fig.1. presents an overview of our method. The above part is the process of our curriculum learning system which is introduced in Section 3.1. The bottom

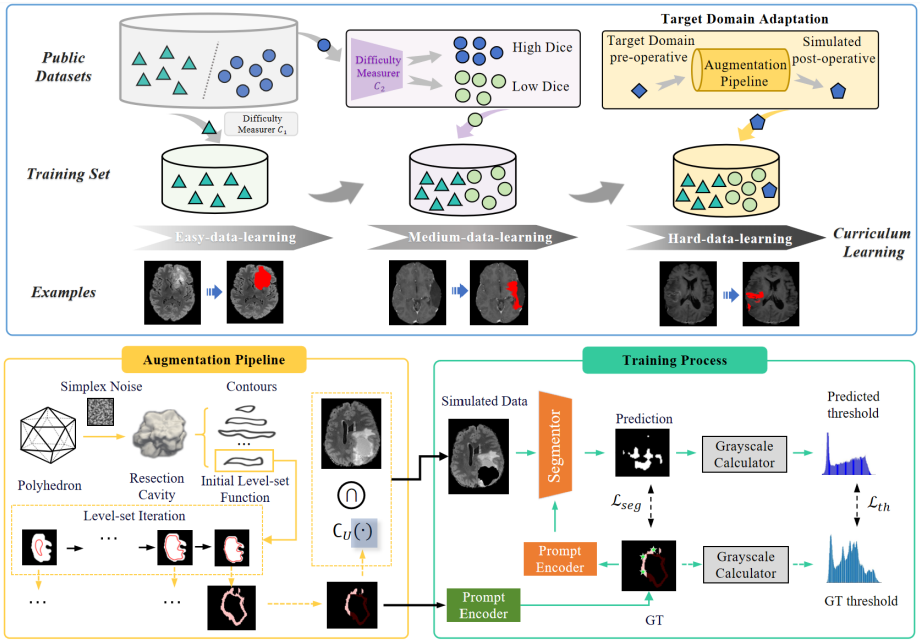



**Fig. 1.** Overview of our framework for postoperative glioma segmentation. The upper part is the process of multi-stage curriculum learning system; the right side of the upper part are examples of each stage's learning case; the lower part is data augmentation pipeline and training process which shows how to simuate postoperative glioma from preoperative and training losses in fine-tune.

part of Fig.1. is the process of postoperative data simulation and training process. The loss function is a weighted sum of segmentation loss and our designed gray-scale threshold loss, which will be detailed in the latter part. A detailed explanation of the simulation process are provided in Section 2.2.

### 2.1 Multi-Stage Curriculum System

Our objective is to achieve precise segmentation of rare and difficult-to-segment post-operative gliomas. Mainstream approaches, such as fine-tuning using datasets with gold standard annotations, have been employed; however, these methods struggle to adapt to the features of target domains sourced from multiple centers, leading to catastrophic forgetting of accumulated knowledge. By designing a learning curriculum, we can assist the model in gradually establishing an understanding of image features, thereby enhancing its generalization ability and recognition accuracy, mitigating overfitting issues, and strengthening the model's adaptability to unseen data.

In this paper, we have designed a multi-stage curriculum based on the framework of Difficulty Measurer + Training Scheduler, and incorporated the difficulty

measurer at different stages as needed to select data in real-time for inclusion in the training set. The Training Scheduler employs the Baby Step Training Scheduler, with the process outlined below. Here we choose $* \in \{1, 2\}$ and $k = 3$ because there are two difficulty measurers and three stages in our method.

**Stage 1: Easy-data Learning** The first step is to engage with easy data. Here, we select a public dataset of primary tumors, which is then processed by a difficulty measure, denoted $C_1$, to determine the data that will be included in the training first phase curriculum. In this context, $C_1$ serves as a volume measurer, selecting primary tumors with smaller volumes to be incorporated into the training set for the first phase. Since primary tumors are generally larger in size and have a relatively regular and uniform shape compared to postoperative gliomas, they pose a lower segmentation difficulty.

**Stage 2: Medium-data Learning** However, due to the properties of post-operative gliomas, such as their blurred boundaries, infiltrative growth, and varying morphologies and sizes, the first-phase model cannot perform well in segmentation. Therefore, new data will be incorporated in the second phase. BRATS2024 is a post-operative glioma dataset with gold standards. Here, we use the model trained in the first phase as a difficulty measurer $C_2$. Data from the BRATS2024 dataset that yield a Dice Similarity Coefficient (DSC) of less than 0.5 when tested through the model training from the previous stage are selected as training data for the second phase and added to the training set for learning.

**Stage 3: Hard-data Learning** After the second phase of training, the model acquires the ability to segment postoperative gliomas. However, due to significant differences between the target domain data and the public dataset in terms of background, tumor volume, and other aspects, target domain adaptation is necessary to equip the model with generalization capabilities in the target domain. Since clinically, postoperative gliomas are difficult to obtain and annotate, there is a scarcity of data with gold standards, making fine-tuning challenging. In contrast, primary tumors are relatively easier to access, and residual tumors are obtained from the resection of primary tumors. Consequently, a two-stage data simulation method is proposed for data augmentation to facilitate fine-tuning, enabling the model to generalize within the target domain. The detailed process will be introduced in section 2.2.

### 2.2 Stage 3: Background Adaptation—Data Simulation

Due to the nature of residual tumors, such as their fuzzy boundaries and infiltrative growth, annotating them is very challenging and time-consuming. However, one thing is certain: residual tumors are obtained from the resection of primary tumors. Consequently, a two-stage data simulation method is proposed. First,

we simulate a noise-distributed polyhedron as the initial contours of each slice; second, using contours obtained from the first step as the initial contours, the level-set method is employed for iteration, with the resected volume controlled to obtain the simulated post-operative residual tumor.

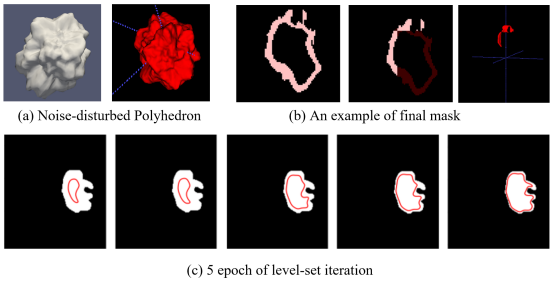



**Fig. 2.** (a) A nosie-distributed polyhedron in vtp format and nifti format. (b) An example of final mask. Left is tumor after level-set iteration, and we radomly cut tumors in several octants. Middle is the mask after cutting. Right is the final mask in nifti format. (c) The showcase of our level-set method iteraiton. The inside part of red contours will be resected in our approach.

**Step I: Noise-disturbed Polyhedron** We start by creating a geodesic polyhedron as a prototype for the initial contours of simulated post-operative tumours. Subsequently, we apply simplex noise to perturb the surface of the polyhedron in order to simulate the surgical resection process and the resulting residual cavity more accurately, introducing a perturbed sphere by radially displacing each vertex. The final volume is obtained through random affine transformations, so the final volume is: $\widetilde{V} = T\left(V, Z_s\right)$

**Step II: Level-set Method** The active contour model, initially proposed by Kass et al.[19] In 1988, image segmentation was transformed into a problem of solving the extremum of an energy function by constructing an energy function. However, early implementation, like the snake model[20], exhibits several limitations, including sensitivity to the initial position, susceptibility to local extrema, inability to converge to concave contours deeply, and the lack of automatic topology transformation capabilities.

We incorporate the Level Set Method, a numerical technique for tracking contour and surface movements, to further control the volume of residual tumors, thereby simulating varying degrees of surgical resection among different physicians. The grayscale information of image pixels is utilized as energy to construct an energy function. This function controls the grayscale values inside and outside the curve, thereby facilitating the evolution of the curve. The equation is eq(4).

Here $I(x, y)$ represents the grayscale value of the original image at the coordinate $(x, y)$, while $c_1$ and $c_2$ denote the mean grayscale values of the image regions inside and outside the curve, respectively. When the curve evolves to the target contour, $Ec$ reaches its minimum value, and energy conservation is maintained both inside and outside. By adding some regularization terms to $Ec$ , an energy functional is obtained in eq(5).

$$Ec = \iint |I(x,y) - c_1|^2 \, dx \, dy \iint |I(x,y) - c_2|^2 \, dx \, dy \tag{1}$$

$$E = \mu \, \text{Length}(C) + \nu \, \text{Area}(C) + Ec \tag{2}$$

$$\frac{\partial}{\partial t} = \delta(\phi) \left[ \mu \, \text{div} \left( \frac{\nabla u}{|\nabla u|} \right) - v - \lambda_1 (I - c_1)^2 \lambda_2 (I - c_2)^2 \right] \tag{3}$$

Therefore, during the iteration process, we can obtain the level set function $\phi$ and energy function $E$ for each iteration. In this paper, to obtain the excision cavity, we initiate from the edge of the foreground portion of the mask and iteratively search for the desired contour $(E)$, which ultimately serves as the boundary of the excision cavity. The Fig.2. show five iterations of one slice of the mask with contours generated by the Level Set Method. Then, we obtained the preliminary mask. However, the level set-based method will retain the outermost tumor, which is inconsistent with the surgical resection strategy. Next, we will perform post-processing on the mask. The post-processing steps are relatively simple. We split each slice of the mask obtained in the previous step area by area and then select the regions we want to set to 0. The remaining areas are the final simulated residual tumors. We have provided the simulated residual tumor data to the neurosurgeon for evaluation, which has met the standard.

### 2.3 Training Process

The training process is shown in Algorithm 1. At each stage, we train our segmentor $f(\bullet)$ as follows: Let input simulated image $X_i$ and label $Y_i$ , the prompt $p = (a,\ b,\ c \mid l)$ ,where $a,\ b,\ c \in 0,\ 1,\ \ldots\ ,\ N - 1$ and $l \in \{0,\ 1\}$, which represents foreground or background, the output is: $Y_o = f(X_i, p)$. In the training process, we employ a segmentation loss in conjunction with a specifically designed gray-scale threshold loss function for supervision. As illustrated below, the segmentation loss is a weighted sum of the standard dice and cross-entropy losses. Here we choose $\lambda = 0.5$ in eq(4).

Moreover, in the third stage of hard data learning, the gray-scale threshold loss we designed calculates the mean squared error of the tumor gray-scale values within the ground truth tumor region and the predicted tumor region based on the initial given prompt points $p_0$. So let grayscale calculator $G(\bullet)$, predicted tumor region $R_o$ and ground truth tumor region $R_i$ calculated in eq(5). The used segmentor is a pre-trained SAM-Med 3D model.

$$L_{seg} = \lambda L_{dice} + (1 - \lambda) L_{entropy} \tag{4}$$

$$L_{th} = MSE\left(\bar{G}\left(R_i\right), \bar{G}\left(R_o\right)\right) \tag{5}$$

**Algorithm 1** MSCL Training Process

**Require:** $D$: Training Dataset; $C_*(* = 1, 2, ....n)$: Difficulty Measurers
**Ensure:** $f^*$: the final model
1: $D' = \text{sort}(D, C_*)$
2: $\{D^1, D^2, \ldots, D^k\} = D'$ where $C_*(d_a) < C_*(d_b)$, $d_a \in D^i$, $d_b \in D^j$, $\forall i < j$
3: $D_{\text{train}} = \emptyset$
4: **for** $s = 1$ **to** $k$ **do**
5: $\quad D_{\text{train}} \leftarrow D_{\text{train}} \cup D$
6: $\quad$ **while** not converged **do**
7: $\quad\quad \text{train}(f, D_{\text{train}})$
8: $\quad$ **end while**
9: **end for**

## 3 Experiments

### 3.1 Datasets and Implementation Details

Brats2022 (1261 cases) and 2024 (1350 cases) are two public datasets. RHUH and ReMIND are two small-sample public datasets, which we use 40 and 29 cases separately. Notably, the ReMIND dataset does not include post-operative data for all patients, and some tumors are not gliomas. Hence, the numbers displayed in the table represent the filtered quantities after such considerations. Moreover, the in-house dataset (60 cases) is from our local hospital.Here, we use pre-operative data to simulate post-operative data for data simulation in the third stage of our curriculumn learning systemand fine-tune the segment with them. Then, post-operative data is for validation.

We used PyTorch for experiments on two Tesla v100 GPUs with an Intel Xeon Gold 6230 processor on Linux. Our backbone was the SAM-Med 3D model. MRI images were preprocessed with normalization, intensity correction, and skull stripping, then transformed and resized to 128×128×128 using TorchIO. Training used a batch size of 1, learning rate of 8e-4, weight decay of 0.1, for 200 epochs.

### 3.2 Results

**Comparison with State-of-the-art Methods** We compared several state-of-the-art methods, including CNN-based, transformer-based, and foundation models, on three datasets. The results are presented in Table 2 below, and our approach outperforms traditional CNN based methods such as U-Net in terms of performance. As to nnU-Net and nnFormer, which are known as solid models, results of our model are also better than those.

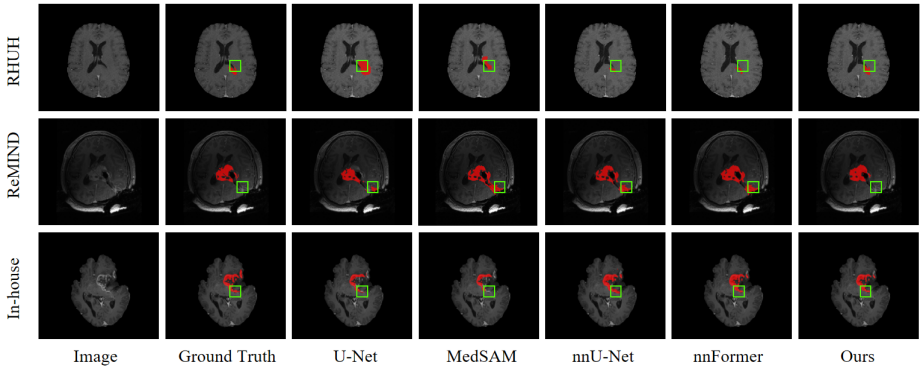



**Fig. 3.** Visualization of different state-of-art methods and our method in three datasets.

**Table 1.** Results of different state-of-art methods with different datasets and metrics reached. DSC represents dice similarity coeeficient, IOU represents intersection over union and HD represents hausdorff distance.

| **Methods** | Brats2024 | | | RHUH | | | ReMIND | | | In-house | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | DSC | IoU | HD | DSC | IoU | HD | DSC | IoU | HD | DSC | IoU | HD |
| U-Net | 51.9 | 40.1 | 23.8 | 24.9 | 17.7 | 56.4 | 48.1 | 35.6 | 39.1 | 50.8 | 34.1 | 26.1 |
| U-Net 3+ | 46.9 | 35.3 | 29.9 | 22.3 | 14.4 | 61.1 | 36.2 | 23.3 | 52.3 | 40.3 | 25.2 | 21.2 |
| DeepResU-Net | 46.5 | 35.2 | 22.9 | 18.7 | 12.9 | 70.3 | 47.2 | 34.4 | 41.8 | 66.4 | 49.7 | 29.9 |
| nnU-Net | 72.8 | 58.9 | 18.6 | 66.0 | 51.4 | 47.3 | 60.9 | 44.6 | 18.9 | 78.3 | **72.8** | 22.8 |
| nnFormer | 72.8 | 58.5 | 18.8 | 68.3 | 53.3 | 43.8 | 49.0 | 34.0 | 38.8 | **84.0** | 72.4 | 21.9 |
| SAM-Med 3D | 73.1 | 57.1 | 16.6 | 49.6 | 33.2 | 15.5 | 67.2 | 51.5 | 19.6 | 69.8 | 53.6 | 16.2 |
| **Ours** | **84.5** | **62.3** | **15.6** | **76.1** | **61.5** | **12.6** | **75.4** | **60.6** | **14.9** | **84.0** | 71.6 | **12.5** |

**Table 2.** Ablation study of critical modules in our proposed method and their segmentation performances.

| Baseline | CL Phase | Threshold Loss | Brats2024 | | | RHUH | | | ReMIND | | | In-house | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | DSC | IoU | HD | DSC | IoU | HD | DSC | IoU | HD | DSC | IoU | HD |
| ✓ | - | - | 60.1 | 43.3 | 18.7 | 37.2 | 19.7 | 35.5 | 45.4 | 29.7 | 49.4 | 64.5 | 47.6 | 28.6 |
| ✓ | - | ✓ | 73.1 | 57.1 | 16.6 | 49.6 | 33.2 | 15.5 | 67.2 | 51.5 | 21.8 | 69.8 | 53.6 | 16.2 |
| ✓ | ✓ | - | 74.4 | 60.5 | 16.9 | 75.4 | 60.1 | 18.3 | 74.3 | 59.1 | 27.0 | 83.4 | 71.5 | 12.8 |
| ✓ | ✓ | ✓ | **84.5** | **62.3** | **15.6** | **76.1** | **61.5** | **12.6** | **75.4** | **60.6** | **27.0** | **84.0** | **71.6** | **12.5** |

**Ablation Study on Modules Employed** We conducted an ablation study on different modules employed in the segmentation task. We explored various combinations of experimental conditions: a) baseline: directly SAM-Med 3D model test; b) using our curriculumn learning strategy; c) gray-scale threshold loss function. The results are obtained in the following table. Based on the results obtained from the ablation experiments, it can be concluded that compared to directly testing on the SAM-Med3d model,using curriculumn learning strategy leads to a big improvement (38.2%, 28.9% and 18.9% in three datasets) in terms of DSC, IoU and HD, indicating that our simluation data is a key factor in

improving segmentation performance. Furthermore, incorporating our designed loss function further enhances the metrics.

## 4 Conclusion

In this paper, we mainly address the issues of limited data and high annotation costs by proposing multi-stage curriculumn learning models that enable the model to engage in continuous learning from preoperative to postoperative data, which preserves the dark knowledge acquired in previous tasks. In the first stage, we choose pre-operative gliomas from pubilc datasets for easy-data-learning; in the second stage, we choose data from the public postoperative dataset that yield a DSC of less than 0.5 when tested through the model training from the previous stage; Ultimately, we undertake target domain adaptation for background knowledge learning, introducing a novel simulation method, which leverages few-shot primary tumors from the target domain to emulate postoperative gliomas using level-set method, enhancing our dataset. We evaluated the proposed method on three public postoperative glioma datasets and one private dataset, achieving improvements in DSC by 11.7%, 15.1%, 8.2%, and 14.2% compared to fine-tuning foundation model directly; 11.4%, 7.8%, 14.5% and 8.7% compared to state-of-the-art methods. These results underscore the effectiveness and robustness of the proposed approach in addressing the challenges associated with post-operative glioma segmentation.

## References

1. Basta, Marguerite B., et al.: Automated Segmentation and Recurrence Risk Prediction of Surgically Resected Lung Tumors with Adaptive Convolutional Neural Networks. arXiv preprint arXiv:2209.08423 (2022)
2. Helland, Ragnhild Holden, et al.: "Segmentation of glioblastomas in early post-operative multi-modal MRI with deep neural networks." Scientific reports 13.1 (2023): 18897.
3. Yao, Yuan, et al. "Unsupervised Domain Adaptation through Shape Modeling for Medical Image Segmentation." International Conference on Medical Imaging with Deep Learning. PMLR (2022).
4. Kirillov, Alexander, et al. "Segment anything." arXiv preprint arXiv:2304.02643 (2023).
5. Wang, Haoyu, et al. "Sam-med3d." arXiv preprint arXiv:2310.15161 (2023).
6. Beiko, Jason, et al. "IDH1 mutant malignant astrocytomas are more amenable to surgical resection and have a survival benefit associated with maximal surgical resection." Neuro-oncology 16.1 (2014): 81-91.
7. Stein, Josh, Maxime Di Folco, and Julia A. Schnabel. "Influence of Prompting Strategies on Segment Anything Model (SAM) for Short-axis Cardiac MRI Segmentation." BVM Workshop. Wiesbaden: Springer Fachmedien Wiesbaden (2024).
8. Mazurowski, Maciej A., et al. "Segment anything model for medical image analysis: an experimental study." Medical Image Analysis 89 (2023): 102918.

9. Zhang, Yichi, Zhenrong Shen, and Rushi Jiao. "Segment anything model for medical image segmentation: Current applications and future directions." Computers in Biology and Medicine (2024): 108238.
10. Ma, Jun, et al. "Segment anything in medical images." Nature Communications 15.1 (2024): 654.
11. Sathish, Rachana, et al. "Task-Driven Prompt Evolution for Foundation Models." International Conference on Medical Image Computing and Computer-Assisted Intervention. Cham: Springer Nature Switzerland (2023).
12. Cheng, Junlong, et al. "Sam-med2d." arXiv preprint arXiv:2308.16184 (2023).
13. Ermiş, Ekin, et al. "Fully automated brain resection cavity delineation for radiation target volume definition in glioblastoma patients using deep learning." Radiation oncology 15.1 (2020): 1-10.
14. Pérez-García, Fernando, et al. "Simulation of brain resection for cavity segmentation using self-supervised and semi-supervised learning." Medical Image Computing and Computer Assisted Intervention–MICCAI 2020: 23rd International Conference, Lima, Peru, October 4–8, 2020, Proceedings, Part III 23. Springer International Publishing (2020).
15. Pérez-García, Fernando, et al. "A self-supervised learning strategy for postoperative brain cavity segmentation simulating resections." International Journal of Computer Assisted Radiology and Surgery 16 (2021): 1653-1661.
16. T. A. Soomro et al., "Image Segmentation for MR Brain Tumor Detection Using Machine Learning: A Review," in IEEE Reviews in Biomedical Engineering, vol. 16, pp. 70-90, (2023).
17. O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation, " in Proc. Medical Image Computing and Computer Assistant Intervention, (2015) pp. 234–241.
18. KASS,M. snake Active contour models[J]. Proc.first Intl Conf.computer Vision, 1988, 4.
19. Caselles V . Geometric models for active contours[C]// International Conference on Image Processing. IEEE Computer Society, 1995.
20. Santiago Cepeda, Sergio García-García, Ignacio Arrese, Francisco Herrero, Trinidad Escudero, Tomás Zamora, Rosario Sarabia, The Río Hortega University Hospital Glioblastoma dataset: A comprehensive collection of pre-operative, early post-operative and recurrence MRI scans (RHUH-GBM), Data in Brief, Volume 50, 2023, 109617, ISSN 2352-3409. https://doi.org/10.1016/j.dib.2023.109617.
21. Juvekar, P., Dorent, R., Kögl, F., Torio, E., Barr, C., Rigolo, L., Galvin, C., Jowkar, N., Kazi, A., Haouchine, N., Cheema, H., Navab, N., Pieper, S., Wells, W. M., Bi, W. L., Golby, A., Frisken, S., & Kapur, T. (2023). The Brain Resection Multimodal Imaging Database (ReMIND) (Version 1) [dataset]. The Cancer Imaging Archive. https://doi.org/10.7937/3RAG-D070.